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Article

Diffusion of Polymethylene Chain Molecules in Nonpolar Solvents

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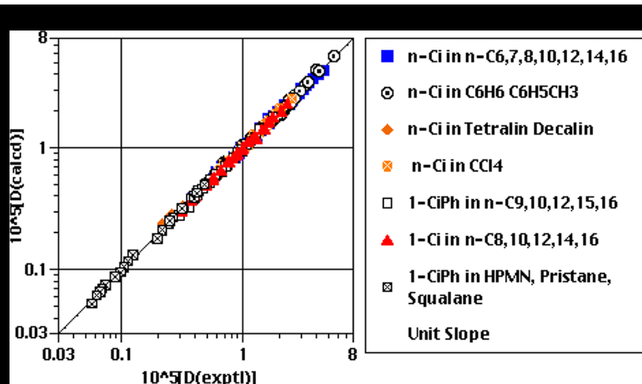
ABSTRACT: [redacted]



■ INTRODUCTION

Polymethylene chain molecules such as the *n*-alkanes are model systems for the study of the hydrodynamic interactions between the fr ional elements of polymers. Kirkwood−Riseman theory takes these interactions into account and has been used in analyses of macromolecular transport properties. Paul and Mazo, in an early test of the theory, considered chains with up to 40 spherical elements (beads) and used Monte Carlo-generated conformers to calculate the size parameter $\phi(r)$ in the Stokes−Einstein relation for the tr lational diffusion constant, D, with stick boundary conditions

$$D = k_B T / 6\pi\eta\phi(\mathrm{r}) \quad (1)$$

where k_B is Boltzmann's constant, T is the absolute temperature, and η is the viscosity; $\phi(r)$ has units of length and is determined by the number of chain elements, their size, and the hydrodynamic interactions. Paul and Mazo's values of $\phi(r)$, calculated with an element radius a = 0.77 Å, gave diffusion constants in agreement with those for *n*-alkanes with 5−32 carbon atoms in CCl_4. The fit was not as good for *n*-alkane solutes in other solvents but could be improved using different values of a that were attributed to solute−solvent interactions.

In a sequent series of papers, Garcia de la Torre and co-workers carried out Kirkwood−Riseman-based calculations of $\phi(r)$ for polymethylene chains using an improved conformation-averaging procedure and a more rigorous method of solving the associated hydrodynamic equations. Their results were consistent with the experimental D values for the *n*-alkan n CCl_4 and b e for a chain element radius of 0.6 Å. Other groups have used Kirkwood−Riseman theory to analyze the diffusion of *n*-alkanes, but the number of solute−solvent systems considered was relatively small. In this paper, the results in refs and are used to determine an average difference of <3% between 207 experimental and calculated diffusion constants that vary by a factor of ∼100, as do the solutions' viscosities.

We find an average difference of <3% for 131 diff i constan f *n*-alkane solutes in seven *n*-alkanes, benzene, toluene, tetralin, decalin, and CCl_4. Similar analyses give average differences of <2% between 26 experimental and calculated D values of 1-alkenes in five *n*-alkanes and <3% for 50 diffusion constants of 1-phenylalkanes in five *n*-alkanes, squalane (2,6,10,15,19,23-hexamethyltetracosane), isocetane (2,2,4,4,6,8,8-heptamethylnonane, MN), and pristane (2,6,10,14-tetramethylpentadecane). The results for the *n*-alkanes and 1-alkenes in the *n*-alkanes are the first from a Kirkwood−Riseman analysis in a homologous series of solvents.

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The analyses in new D values, determined using capillary flow techniques, ' for 1-alkenes (1-hexadecene and 1-eicosene in n-hexane, n-octane, n-decane, n-dodecane, n-tetradecane, n-hexadecane, and squalane) and 1-phenylalkanes (ethylbenzene, 1-phenylpentane, 1-phenyloctane, 1-phenylundecane, 1-phenyltetradecane, and 1-phenylheptadecane in n-hexadecane). Five of these 20 diffusion constants could not be compared with calculated values for reasons explained in the " " section.

Our average difference for the bead model is comparable to those obtained for many of the same solute−s t systems using the cylinder and lollipop diffusion models. ' The three models and their results are compared.

■ EXPERIMENTAL METHODS

Chemicals and Sample Preparation. In this and the following sections, n-C$_i$, 1-C$_i$, and 1-C$_i$Ph are used for particular n-alkanes, 1-alkenes, and 1-phenylalkanes, respectively. The suppliers for our solutes and solvents were (a) Aldrich: 1-C$_8$Ph (98%), 1-C$_{11}$Ph (99%), 1-C$_{17}$Ph (97%), n-C$_8$ (99+%), n-C$_{14}$ (99+%), and squalane (99%); (b) Sigma-Aldrich: n-C$_6$ (≥99%), n-C$_{10}$ (≥99%), and n-C$_{16}$ (99%); (c) Alfa Aesar: 1-C$_5$Ph (96%) and 1-C$_{14}$Ph (97%); (d) TCI: 1-C$_{16}$ (>99%) and 1-C$_{20}$ (>97%); (e) Sigma: n-C$_{12}$ (99%); and (f) Acros: ethylbenzene (99.8%). All substances were used as received. Solutions were prepared by adding two to five drops of the liquid solutes to 6.0−8.0 mL of the solvents.

Data Acquisition and Analysis. The microcapillary method used to determine the D values was developed by Bello and co-workers. The acquisition of the solutes' elution profiles was described in refs and . The data acquisition system, including the microcapillary (76.5 μm i.d.) and variable wavelength detector, was the same as that in ref ; wavelengths of 260 and 192 nm were used for the 1-phenylalkanes and 1-alkenes, respectively. The diffusion constants were obtained by comparing the sigmoidal elution profiles with those calculated using Taylor's theory, as described in refs , , and . All profiles were taken at room temperature, which was 22.50 ± 0.50 and 25.00 ± 0.25 °C for the 1-alkenes and 1-phenylalkanes, respectively. Multiple determinations of D were made for each solute−solvent pair; their average values, with uncertainties of ±2.5%, are given in and .

Table 1. Diffusion Constants of 1-C$_{16}$ and 1-C$_{20}$ in Alkanes at 22.50 ± 0.50 °C

	10^6D(expt), cm^2 s^{-1}	
solvent	1-C$_{16}$	1-C$_{20}$
n-C$_6$	24.0	20.1
n-C$_8$	15.0$_5$	13.0
n-C$_{10}$	9.81	8.18
n-C$_{12}$	6.61	5.67
n-C$_{14}$	4.94	4.04
n-C$_{16}$	3.75	3.08$_5$
squalane	0.876	0.667

Table 2. Diffusion Constants of Alkylbenzenes in n-C$_{16}$ at 25.0 ± 0.25 °C

solute	10^6D(expt), cm^2 s^{-1}
ethylbenzene	7.76$_5$
1-C$_5$Ph	5.91
1-C$_8$Ph	4.60
1-C$_{11}$Ph	3.78
1-C$_{14}$Ph	3.24
1-C$_{17}$Ph	2.95

Sol Solvent Systems. The solv for the n-alkane, − 1-alkene, and 1-phenylalkane ' solutes are given in − , respectively, as are the temperatures at which they were studied. A temperature near (or equa °C was used for all solutions; the solvents' viscosities ' − at those temperatures are given in − and will be used when discussing our results in the sections " " and " ". The n-alkanes in toluene, CCl$_4$, and benzene ( ) and the 1-alkenes in the n-alkanes ( ) were studied ional temperatures; the viscosities at those temperatures ' − are not given but were used in our analyses. The diffusion constants in refs − were reported from two to four significant figures; the average differences between the experimental and calculated D values reported here are given to two significant figures with a third subscripted figure.

■ RESULTS AND DISCUSSION

Hyd ynamic Bead Model. Garcia de la Torre and co-workers ' calculated values of $\phi(r) = f_t^*$ averaged over Monte Carlo-generated polymethylene conformers at 25 °C with a 500 cal mol^{-1} energy difference between t and g rotational isomers. Hydrodynamic interactions were taken into account using a modified Oseen tensor; translation−rotation coupling was included as was excluded volume. Stokes law

$$f_i = 6\pi\eta\sigma \tag{2}$$

was used as the friction coefficient for the N spherical chain elements (beads) with a common hydrodynamic radius, σ, and a separation b = 1.53 Å between elements. Values of f_t^* were report s a function of σ/b for $0.0654 \leq \sigma/b \leq 0.490$ and $50 \geq N \geq 5$. ' We calculated diffusion constants using values of f_t^* interpolated from Table 2 of ref and Figure 3 of ref ; σ/b was varied to find the smallest average difference between the experimental and calculated D values for a given type of solute in a given solvent. The best values of σ/b and σ were solvent-dependent and are given in , , and for the n-alkanes, 1-alkenes, and 1-phenylalkanes, respectively; the corresponding average differences are given in .

The agreement for the n-alkane solutes' 131 experimental and calculated diffusion constants ($32 \geq N \geq 5$) was generally good with an average difference of 2.8$_1$%. The average difference was 2.6$_0$% in the n-alkanes, 3.2$_3$% in the other hydrocarbon solvents (benzene, toluene, tetralin, and decalin), and 2.2$_9$% in CCl$_4$ ( ). Our values of σ = 0.661 Å for CCl$_4$ and 0.623 Å for benzene ( ) are consistent with $\sigma \sim 0.6$ Å found for both solvents by Garcia de la Torre and co-w ers (who did not vary σ/b to the same extent as this analysis). '

The n-alk diffusion constants in toluene, CCl$_4$, ' ' and benzene ' were determined over ranges of 73.0, 43.5, and 2.2 °C, respectively, but temperature-dependent values of σ/b were not needed to produce acceptable agreement with experiment. We found average differences of 2.2$_9$% in CCl$_4$, 1.9$_7$% in toluene, and 4.1$_6$% in benzene ( ) using a single value of σ/b for each solvent ( ). Paul and Mazo's values of $\phi(r)$ shown to be insensitive to temperature for n-alkane solutes; ' ' that appears to be the case for f_t^* as well.

Table 3. *n*-Alkane Solutes and Their Solvents

solvent	i for n-C$_i$ solutes	T (°C)
n-C$_6$	32, 24, 18, 16, 12, 10, 8, 7, 6, 5	25.0
n-C$_7$	18, 16, 14, 12, 10, 8, 7, 6	25.0
n-C$_8$	32, 24, 16, 14, 12, 8	25.0
n-C$_{10}$	16, 7	25.0
n-C$_{12}$	18, 16, 12, 8, 7, 6	25.0
n-C$_{14}$	8, 7	25.0
n-C$_{16}$	16, 12, 10, 8, 7, 6	25.0
benzene	32, 29, 28, 18, 14, 12, 10, 9, 8, 7, 6, 5	22.2
benzene	20, 18, 14, 10, 5	20.0
toluene	26, 22, 18, 14, 10, 6	27.0, 50,0, 75.0, 100.0
tetralin	28, 18, 14, 12, 9, 7, 6, 5	22.2
decalin	28, 18, 14, 12, 10, 9, 8, 7, 6, 5	22.2
CCl$_4$	32, 28, 22, 20, 18, 16, 12, 10, 8, 7, 6, 5	25.0
CCl$_4$	26, 18, 14, 10, 6	20.5, 35.0, 50.0, 64.0

[a]The D value for 6 is the average of the values from ref ; the rest are from ref . [b]The D value for 18 is from ref ; that for 16 is from ref ; those for 14, 12, and 10 are from ref ; that for 8 is from ref ; that for 7 is from ref ; and that for 6 is the average of those from ref . [c]The D value for 8 is the average of the values in refs – ; that for 12 is from ref ; that for 14 is from ref ; and those for 16, 24, and 32 are from ref . [d]The D value for 7 is from ref ; the D value for 16 is from ref . [e]The D value for 6 is from 22, that for 7 is from ref , those for 8 and 18 are from ref , that for 12 is the average of the values in refs and , and the value for 16 is from ref . [f]The D values for 7 and 8 are from ref . [g]The D values for 6, 7, and 8 are from ref ; those for 10 and 12 are from ref ; and that for 16 is the average of the values in refs and . [h]All D values are from ref . [i]All D values are from ref . [j]All D values are from ref . [k]The D values for 28 and 32 are from ref ; all others are from ref .

Table 4. 1-Alkene Solutes and Their Solvents

solvent	i for 1-C$_i$ solutes	T (°C)
n-C$_8$	14, 12, 10, 8	25.0
n-C$_8$	20, 16	22.5
n-C$_{10}$	14, 12, 10, 8	25.0
n-C$_{10}$	20, 16	22.5
n-C$_{12}$	14, 12, 10, 8	25.0
n-C$_{12}$	20, 16	22.5
n-C$_{14}$	14, 12, 10, 8	25.0
n-C$_{14}$	20, 16	22.5
n-C$_{16}$	20, 16	22.5

[a]The D values for 1-C$_{16}$ and 1-C$_{20}$ are from this work; the others were calculated using the data in ref .

Table 5. 1-Phenylalkane Solutes and Their Solvents

solvent	i for 1-C$_i$Ph solutes	T (°C)
n-C$_9$	17, 14, 11, 8, 5	24.0
n-C$_{10}$	17, 14, 11, 8, 5	23.5
n-C$_{12}$	17, 14, 11, 8, 5	20.5
n-C$_{15}$	17, 14, 13, 12, 11, 9, 8, 7, 6, 5	24.5
n-C$_{16}$	17, 14, 11, 8, 5	25.0
HPMN	17, 14, 11, 8, 5	21.0
pristane	17, 14, 11, 8, 5	25.5
squalane	17, 14, 13, 12, 11, 9, 8, 7, 6, 5	23.0

[a]The D values in n-C$_{16}$ are from this work; those in n-C$_9$, n-C$_{10}$, and n-C$_{12}$ are from ref ; and those in n-C$_{15}$, HPMN, pristane, and squalane are from ref .

While the terminal groups of the 1-alkenes and 1-phenylalkanes are obviously different from the n-alkanes' methyl groups, we found an average difference of 1.8$_8$% for these solutes' 76 diffusion constants by assuming that the 1-phenylalkanes' phenyl ring and the 1-alkenes' double bond added a single equivalent element to their chains of n_c sp^3 carbon atoms ($N = n_c + 1$); solutes with $n_c \geq 5$ were considered with $17 \geq n_c \geq 5$ for the 1-phenylalkanes and $18 \geq n_c \geq 6$ for the 1-

Table 6. Values of the Solvents' Viscosities, σ/b, σ, and c_β for *n*-Alkane Solutes

solvent	T (°C)	$10^2\eta$ (P)	σ/b	σ (Å)	c_β
n-C$_6$	25.0	0.300	0.464	0.710	
n-C$_7$	25.0	0.385	0.429	0.656	
n-C$_8$	25.0	0.508	0.380	0.581	
n-C$_{10}$	25.0	0.838	0.274	0.419	
n-C$_{12}$	25.0	1.38	0.196	0.300	
n-C$_{14}$	25.0	2.09	0.163	0.249	
n-C$_{16}$	25.0	3.03	0.135	0.207	
toluene	27.0	0.535	0.422	0.646	
benzene	22.2	0.632	0.407	0.623	0.213
tetralin	22.2	2.10	0.222	0.340	0.121
decalin	22.2	2.64	0.196	0.300	0.099
CCl$_4$	25.0	0.894	0.432	0.661	0.225

[a]See for the list of solutes. [b]If the D values were analyzed at more than one temperature in a given solvent (see), the temperature closest to (if not at) 25 °C and the corresponding viscosity are given. [c]Values of σ for b = 1.53 Å. [d]From ref for benzene and CCl$_4$ at 25 °C, decalin and tetralin at 22.2 °C. [e]From ref . [f]Viscosities for toluene, benzene, and CCl$_4$ at all temperatures in are from ref . [g]From ref .

Table 7. Values of the Solvents' Viscosities, σ/b, and σ for 1-Alkene Solutes

solvent	T (°C)	$10^2\eta$ (P)	σ/b	σ (Å)
n-C$_8$	25.0	0.508	0.419	0.641
n-C$_{10}$	25.0	0.838	0.295	0.451
n-C$_{12}$	25.0	1.38	0.221	0.338
n-C$_{14}$	25.0	2.09	0.173	0.265
n-C$_{16}$	25.0	3.03	0.147	0.225

[a]See for the list of solutes. [b]The viscosities at 25 °C are given, although some of the D values were obtained at 22.5 °C (see); the viscosities for both temperatures are from ref . [c]Values of σ for b = 1.53 Å.

Table 8. Values of the Solvents' Viscosities, σ/b, and σ for 1-Phenylalkane Solutes

solvent	T (°C)	$10^2\eta$ (P)	σ/b	σ (Å)	1-C_iPh/n-C_i	1-C_iPh/1-C_i
n-C_9	24.0	0.674	0.459	0.702		
n-C_{10}	23.5	0.857	0.453	0.693	1.65	1.54
n-C_{12}	20.5	1.50	0.353	0.540	1.80	1.60
n-C_{15}	24.5	2.57	0.217	0.332		
n-C_{16}	25.0	3.03	0.216	0.330_5	1.60	1.47
HPMN	21.0	3.44	0.234	0.358		
pristane	25.5	6.61	0.114	0.174		
squalane	23.0	30.3	0.0717	0.110		

[a]See for the list of solutes. [b]Values of σ for b = 1.53 Å. [c]σ(1-C_iPh)/σ(n-C_i). [d]σ(1-C_iPh)/σ(1-C_i). [e]From ref . [f]From ref . [g]From ref . [h]From ref .

Table 9. Values of Average Percentage Difference

solutes	no. of solutes	solvents	avg % diff
n-C_i	40	n-C_i	2.6_0
n-C_i	8	tetralin	3.7_9
n-C_i	10	decalin	4.2_4
n-C_i	17	benzene	4.1_6
n-C_i	24	toluene	1.9_7
n-C_i	59	te, de, bz, to	3.2_3
n-C_i	32	CCl_4	2.2_9
n-C_i	131	total	2.8_1
1-C_i	26	n-C_i	1.5_0
1-C_iPh	30	n-C_i	1.7_0
1-C_iPh	20	H, P, S	2.6_6
1-C_iPh	50	total	2.0_8
1-C_i, 1-C_iPh	total	76	1.8_8
all solutes	207	all solvents	2.4_7

[a]From – for n-C_i, 1-C_i, and 1-C_iPh, respectively. [b]Tetralin, decalin, benzene, and toluene. [c]HPMN, pristane, and squalane.

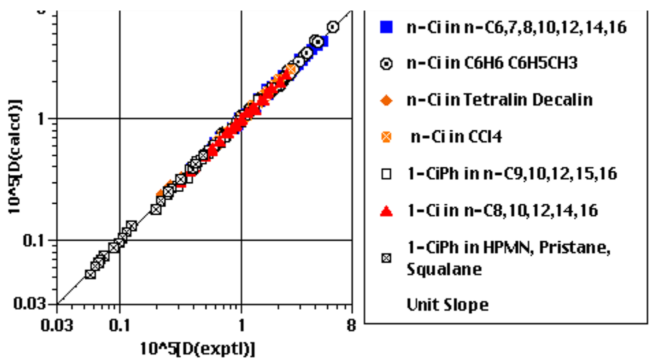

Figure 1. Comparison of the experimental diffusion constants (in cm^2 s^{-1}) for n-alkane, 1-alkene, and 1-phenylalkane solutes with the predictions of a hydrodynamic bead model based on Kirkwood–Riseman theory.

alkenes. The average differences were 2.0_8% for the 1-phenylalkanes in the n-alkanes, HPMN, pristane, and squalane and 1.5_0% for the 1-alkenes in the n-alkanes ( ); the 1-alkenes were studied at two temperatures ( ), and as for the n-alkanes, a single value of σ/b gave satisfactory agreement in each solvent ( ).

This degree of agreement would seem to indicate that, for this model and these solutes, the differences in the shapes and sizes of their terminal groups and the attendant modifications of the conformational averaging are not as important as the number of elements and their nonpolar character (as well as that of the solvents). The 2.4_7% average difference between the 207 experimental and calculated diffusion constants for the n-alkanes, 1-alkenes, and 1-phenylalkanes (and) indicates the model's usefulness for solutions in which the D values and viscosities vary from 5.57×10^{-5} cm^2 s^{-1} for n-C_6 in toluene at 100 °C (η = 0.280 cP) to 5.68×10^{-7} cm^2 s^{-1} for 1-C_{17}Ph in squalane at 23 °C (η = 30.3 cP). It would be interesting to see what differences are found when calculations are carried out for alkyl chains with terminal phenyl and vinyl groups.

The values of σ/b for our fits determine the strength of the hydrodynamic interactions between the beads, which is given by

$$\lambda = f_i/8\pi\eta b \quad (3)$$

which becomes

$$\lambda = (3/4)(\sigma/b) \quad (4)$$

when is used for f_i. Zwanzig, Kiefer, and Weiss, in an analysis of Kirkwood–Riseman theory, said that "interesting values of λ can range from 0.10 to 10". While λ = 0.10 was not stated to be an absolute lower limit, it gives σ/b = 0.133, which is greater than the values for the 1-phenylalkanes in pristane (0.114) and squalane (0.0717) and is close to that for the n-alkanes in n-C_{16} (0.135).

A reviewer questioned whether it was appropriate to use a hydrodynamic model for these small values of σ/b and addressed the issue by calculating the self-diffusion constants for n-C_8, n-C_{12}, and n-C_{16}. These are solvents for which our n-alkane solutes, including their self-diffusion constants, had values of σ/b that decreased from 0.380 for n-C_8 to 0.196 for n-C_{12} to 0.135 for n-C_{16}. The reviewer's D values for the three n-alkanes were obtained from simulations in boxes of varying length. The diffusion constants for each of the n-alkanes increased as the box length increased. The D values in each box for a given n-alkane became equal; however, if Yeh and Hummer's hydrodynamically based correction to infinite box size was applied, inferring that the molecules' diffusion can be described by classical hydrodynamics and the Kirkwood–Riseman equation can be used for our solvents with higher viscosities and smaller values of σ/b.

Calculations of f_t* for more values of σ/b than those given in refs and also would be helpful because the minimum average differences could not be determined for the 1-alkenes in n-C_6 and squalane. In n-C_6, values of σ/b greater than the upper limit of 0.490 are required for the 1-C_i with i = 8, 10, 12, and 14 from ref and i = 16 and 20 from . In squalane, values less than the lower limit of 0.0654 are needed for the 1-C_i with i = 14 from ref and i = 16 and 20 from , although additional simulations would seem necessary to check if values of σ/b <

0.0654 can be used. The D values for 1-C_{16}, 1-C_{20}, and ethylbenzene (with $n_c = 2$) were reported, since they may be useful in other analyses.

Solvent Dependence of σ. – give our values of σ and, as explained in the " " section, the solvents' viscosities at the temperatures close to (if not at) 25 °C. The values of σ for a given type of solute decrease as the viscosity of a given type of solvent increases; for example, σ = 0.710, 0.419, and 0.207 Å for the n-alkanes in n-C_6, n-C_{10}, and n-C_{16}, respectively. The values of σ also are small compared to 2.00 Å, the common van der Waals radius of the methyl and methylene groups; the largest is σ = 0.710 Å for the n-alkanes in n-C_6, and the smallest is 0.110 Å for the 1-phenylalkanes in squalane.

The results for the n-alkane and 1-alkene solutes in the n-alkanes are, to our knowledge, the first from a Kirkwood–Riseman-based study in a homologous series of solvents; the values of σ can be fitted to

$$\sigma = A_\sigma/\eta^p \tag{5}$$

where η is in cP at 25 °C ( ). The fits, shown in , give $A_\sigma = 0.377_5$ Å, $p = 0.559$, and $R^2 = 0.995$ for the n-alkane

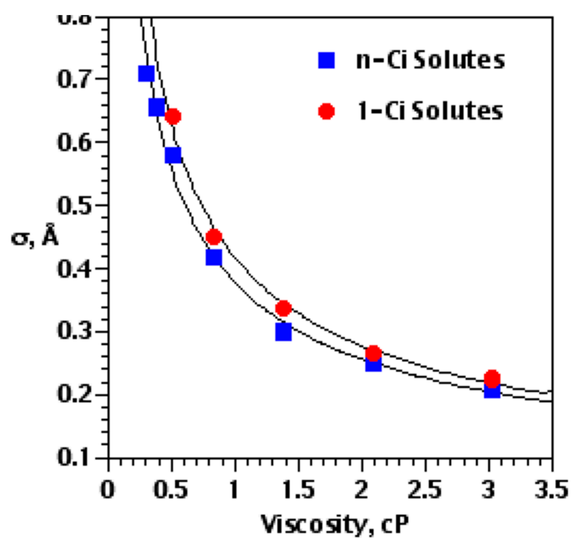

Figure 2. Values of σ for the n-alkane and 1-alkene solutes in the n-alkane solvents vs the solvents' viscosities. The fit is to .

solutes and A_σ = 0.417 Å, p = 0.588, and R^2 = 0.995 for the 1-alkenes; the uncertainties in σ are ±3.5%. The 1-phenylalkanes also are solutes in a homologous series of the n-C_i (i = 9, 10, 12, 15, 16), but the ranges of σ and η are narrower than those for the n-alkanes and 1-alkenes. The n-alkanes in benzene, toluene, decalin, and tetralin are not in a homologous series nor are the 1-phenylalkanes in HPMN, pristane, and squalane, but as seen in , where all three types of solutes in the hydrocarbon solvents are included, the trend is obvious; σ decreases by a factor of ∼6.5, while the viscosities increase by a factor of ∼100.

Stepto and co-workers found a parameter similar to σ in their analysis of the diffusion constants of n-alkanes in four of the solvents considered here: benzene and CCl_4 at 25.0 °C and decalin and tetralin at 22.2 °C. Kirkwood–Riseman calculations were carried out using the rotational-isomeric-state model and a segment f tion constant for the solute suggested by Freire and Horta

$$f_i = 6\pi\eta c_\beta \beta \tag{6}$$

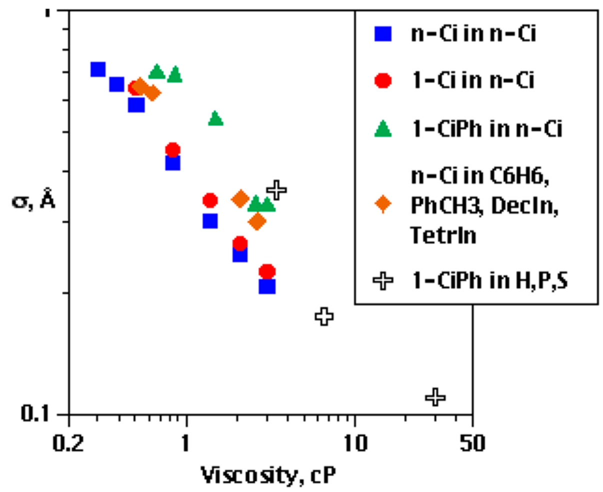

Figure 3. Comparison of the values of σ that give the best agreement between experimental and calculated D values and the solvents' viscosities for n-alkanes in n-alkanes; 1-alkenes in n-alkanes; n-alkanes in benzene, toluene, tetralin, and decalin; 1-phenylalkanes in n-alkanes; and 1-phenylalkanes in HPMN, pristane, and squalane.

where $\beta = (\langle r^2\rangle/n)^{1/2}$ is the effective bond length of a polymethylene chain with n bonds and a root-mean-square end-to-end distance $\langle r^2\rangle^{1/2}$; $c_\beta\beta$ is the effective segment radius. The experimental diffusion constants for the n-alkanes could be fitted with solvent-dependent values of c_β that (with the exception CCl_4 and benzene) decreased as the viscosity increased; they are given in . shows a

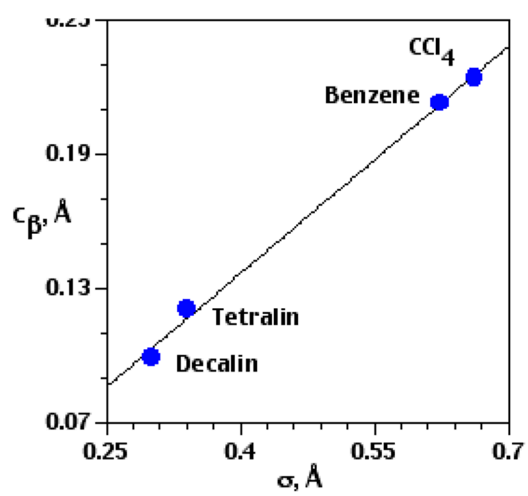

Figure 4. Comparison of the values of c_β from ref with our values of σ from .

reasonably good correlation between c_β and σ; our values of σ for CCl_4 and benzene have the same reversed viscosity dependence as c_β ( ). There is a difference between Stepto's approach and ours; the segmental c n constant in depends on chain length (through β), ile Garcia de la Torre's, , does not. Note that Stepto's parameters were c_b and b; we have used c_β and β to avoid confusion with our b = 1.53 Å.

COMPARISON OF MODELS

Cylinder and Lollipop Models. The diffusion constants for many of the solute–solvent systems considered here also have been compared with the predictions of the cylinder and lollipop models. The solute size parameters from those analyses likewise decreased as the solvents' viscosities increased and were small compared to the van der Waals radius of the methyl and methylene groups. The cylinder model assumed the solute to be a rigid rod divided into elements with diameter d and length l. The experimental diffusion constants were compared with values calculated as a function of l/d. For each value of l/d, the value of d that gave the best agreement with experiment for a given type of solute in a given solvent was determined. A value of $l/d = 1.57$ gave the lowest average difference of 3.3_8% between 150 experimental and calculated D values. The values of d generally decreased as the viscosity increased for n-alkanes in n-alkanes; 1-alkenes in n-alkanes; 1-phenylalkanes in n-alkanes; and 1-phenylalkanes in HPMN, pristane, and squalane.

The diffusion constants of 1-phenylalkanes in n-alkanes, HPMN, pristane, and squalane also were analyzed using the lollipop model. The phenyl group is the candy, a sphere with radius σ_1, and the alkyl chain is the rigid handle, a linear array of spheres with radius σ_2. For each value of σ_1/σ_2, the value of σ_2 that gave the best agreement with experiment for a given type of solute in a given solvent was determined. A value of $\sigma_1/\sigma_2 = 2.17$ gave the lowest average difference of 3.7_3% between 85 experimental and calculated D values. The values of σ_2 that gave the best agreement in the n-alkanes decreased as the viscosity increased, as did those in in HPMN, pristane, and squalane.

The average differences for the bead, cylinder, and lollipop models are comparable, although that for the bead model, which is more characteristic of the solutes, is slightly smaller and was obtained using more D values. The bead model also varied only one size parameter, σ (b remained fixed), while the cylinder model varied l/d and d and the lollipop model varied σ_1/σ_2 and σ_2. One of the conclusions from the cylinder and lollipop analyses, however, was that the reasonably good agreement indicated that the alkyl chains of the n- es, 1-alkenes, and 1-phenylalkanes were relatively extended. ' This was consistent with Monte Carlo and molecu namics (MD) simulations as well as experimental results; − the time-averaged shapes calculated for the n-C$_i$ with i = 8, 12, 16, and 20 were "basically rodlike" with "little change in the length between dynamic time steps". The bead model was intended for n-alkane solutes, and its success with them indicates the values of f_t* are consistent with an extended shape; this may be a reason the model works as well as it does for the 1-alkenes and 1-phenylalkanes.

Solvent-Dependent Size Parameters. The solvent-dependent values of σ, like the solutes' size parameters for the cylinder and lollipop models, are indicative of solute−solvent interactions. Zwanzig and Harrison pointed out that these effective hydrodynamic sizes are a measure of the coupling of the solute motion to the solvent flow; the weaker the coupling, the smaller the value of σ. Two trends in our values of σ, which are most clearly seen in the n-alkane solvents, are consistent with this explanation. The first is the strength of the solutes' interactions with a common solvent. It would be expected to decrease in the order 1-phenlylalkanes > 1-alkenes > n-alkanes because of the differences in their structures. shows this to be the case in n-C$_{10}$, n-C$_{12}$, and n-C$_{16}$; the values of σ for the 1-phenylalkanes are 47−60% larger than those for the 1-alkenes and 60−80% larger than those for the n-alkanes.

The second trend is the decrease in σ as the viscosity increases. Solutes moving through the n-alkanes' relatively extended regions have relatively weak interactions with them. The lengths of these regions increase as their chain lengths (and viscosities) increase, leading to longer intervals of weak interactions and smaller values of σ ( ). As seen in , the correlation is evident although not as strong for the other solute−solvent groupings. Molecular dynamics simulations would be helpful in gaining a more detailed understanding of the relation between the values of σ and the intermolecular forces in these systems.

Others, including Paul and Mazo, have discussed solvent-dependent size parameters in terms of solute−solvent interactions. Stepto and co-workers interpreted their values of c_β in terms of the solutes' interactions with the actual structure of the s s as opposed to their assumed continuous nature. ' ' Garcia de la Torre et al. pointed out that both local structure (short-range) and excluded volume (long-range) interactions contribute to the hydrodynamic properties of real chains.

SUMMARY AND CONCLUSIONS

The diffusion constants of 1-C$_{16}$ and 1-C$_{20}$ in a series of n-alkanes and several 1-phenylalkanes in n-C$_{16}$ have been determined at room temperature using capillary flow techniques. They are among the 207 D values that w compared with the predictions of a hydro amic bead model ' based on Kirkwood−Riseman theory. ' The solutes and solvents, whose D values and viscosities both vary by a factor of ∼100, include (a) n-alkanes in n-alkanes; (b) n-alkanes in benzene, toluene, tetralin, decalin, and CCl_4; (c) 1-alkenes in n-alkanes; (d) 1-phenylalkanes in n-alkanes; and (e) 1-phenylalkanes in methyl-substituted alkanes. The agreement is generally good; the average difference between the experimental and calculated D values is 2.4_7%. This result is comparable to that for many of the same solute−s t systems using the cylinder and lollipop diffusion models. '

The diffusion constants were calculated as a function of the bead radius, which was found to decrease as the solvent viscosity increased; this correlation of solute size parameters with viscosity was also found for many of the same solute−solvent combinations using the cylinder and lollipop models. The solvent dependence of the size parameters is attributed to solute−solvent interactions. The results for the 1-alkenes and n-alkanes in the n-alkanes are the first from a Kirkwood−Riseman analysis in a homologous series of solvents.

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Notes

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